

Web Application Security Assessment Report

Project: Web Application Security Lab

Application: OWASP Juice Shop

Assessment Type: Manual Web Application Security Testing

Tester: Mohammed Marvan

Executive Summary

A security assessment was conducted against the OWASP Juice Shop application in a controlled laboratory environment. The objective was to identify common web application vulnerabilities based on the OWASP Top 10 and document findings with remediation recommendations.

Scope

Included: Authentication, Search functionality, User input fields, HTTP requests, REST API, Client-side validation.

Excluded: Denial of Service, Third-party services, Production infrastructure.

Methodology

Information Gathering → Proxy Configuration → HTTP Request Analysis → Vulnerability Testing → Verification → Documentation.

Tools Used

Burp Suite Community Edition, Firefox, Kali Linux, Docker, FFUF, SQLMap.

Findings

1. Cross-Site Scripting (Medium): Validate input, encode output, implement CSP.
2. SQL Injection Testing (High - Test Scenario): Use parameterized queries and input validation.
3. Broken Authentication (Medium): Strong passwords, rate limiting, secure session management.
4. Directory Enumeration (Low): Disable directory listing and restrict sensitive paths.
5. Security Misconfiguration (Medium): Remove defaults, configure security headers, keep software updated.

Risk Summary

Critical: 0, High: 1, Medium: 3, Low: 1, Informational: 0.

Conclusion

This lab demonstrates practical web application security testing and professional vulnerability documentation aligned with the OWASP Top 10.

Disclaimer

Performed only against OWASP Juice Shop in a controlled lab environment for educational purposes.